

Data Literacy WS2022

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09/02/2023

1 General Probability Theory

We had a grid mesh ($x = [-4, 4]$ and $y = [-4, 4]$) where squares shaded in yellow represented places where it is possible to find the golden treasure. There were 20 yellow squares and 44 other squares (colored in blue). The probability of finding the treasure in yellow squares was uniform.

1. Plot the marginal probability distribution $p(x_G)$
2. What is $p(-1 < x_G < 0 | y = 1.2)$?
3. Are x and y independent?

2 Variance Estimator

We had 4 groups of scientists of the same size. They measured the temperature of the lake Constance. In the first and the second group each scientist took 4 samples and in the third and the fourth group each of them took 25 samples. All the samples are considered independent. Each group estimated the variance of the temperature.

1st group: $\tilde{\sigma} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2$, each scientist took 4 samples

2nd group: $\tilde{\sigma} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \hat{\mu})^2$, each scientist took 4 samples

3rd group: $\tilde{\sigma} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2$, each scientist took 25 samples

4th group: $\tilde{\sigma} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \hat{\mu})^2$, each scientist took 25 samples

We were given 4 different histograms of variance estimates with true variance and estimated variance. We had to match the formulas for estimators with plots.

Idea for solution: Pay attention if the estimator is biased and the standard error
b) Prove that the expected value of the uniform distribution $U(0, \theta)$ with some θ is $\frac{\theta}{2}$.

c) Prove that the estimator for the mean value (can't remember which one) of the uniform distribution $\hat{\phi} = \frac{1}{N}$ is unbiased.

3 Null Hypothesis Testing

We had two groups of people - spinach eaters and non-spinach eaters. The goal of analysis was to see whether people that eat spinach had higher life expectancy. Mean values of life expectancies were given (μ_1, μ_2) , as well as variance which was the same for both groups (σ^2). Significance threshold for p-value was 0.05 and we got p-value 0.01.

1. What would be the reasonable null hypothesis?
2. Mark which statements are true and for those that are false explain why.
 - a) If the experiment was repeated 1000 times, we would get significant result in 990 times.
 - b) We have disproved null hypothesis.
 - c) We can reject null hypothesis.
 - d) The probability that the null hypothesis is false is 0.99
3. We were given the exact values of μ_1 , μ_2 and σ^2 and a z-table relevant for this part of the task. Then we were asked to find the probability of people over 80 years to be spinach-eating.

4 Causal Inference

There are two ways to get admitted to a college. Either through an excellent entrance test (prevalence 37.5%) or through a sports scholarship (prevalence 20%). We assume that everyone who was offered admission has accepted it and that the academic ability and athletic ability are independent.

1. Prove that 50% of the applicants have been admitted.
2. Prove that 75% of the students that were admitted had high score on the entrance test and that 40% had got sports scholarship.
3. Find $Cov(A, Z|C)$. (can't remember the rest)

5 Linear Regression

We were given the dataset of N samples that consisted of the amount of watering and pesticides used for aubergine cultivation (features) and with label that was the weight of aubergine (N of them).

1. Which assumptions about the noise/residuals do we need to make in order to predict the weight from the two features mentioned?
2. Prove that minimizing the summed squared error yields the same result as MLE.
3. If we use the Bayesian inference, what additional assumption would we have to make?

6 PCA

1. What is PCA and give two applications of it.
2. We were given a scatter plot with two features (measurements from two acceleration sensors mounted on the joints of the runners) that were correlated. The task was to approximately draw two principal components.
3. Scientist decided to add ten more sensors. We were given the plot describing the fraction of variances captured. How would you decide about the number of principal components to keep? Give three approaches.
4. Can't exactly remember, but it was something like suggest the analysis that they could perform with the data from the sensors.